engine, and the third stage a cryogenic engine called HM-7, above which the 1-tonne payload would be integrated.

All these went for a toss when the scientists realised that the payload would weigh 1.5 tonnes, and not 1 tonne as originally planned. Now this meant enhancing the boosters, reconfiguring the vehicle. Many high-end rocket engines are upgraded versions of smaller engines, but beyond a point upgrading becomes a misnomer as the new product would virtually be a new engine, though with the basic concepts of its predecessors. In fact, M-40 is such an upgraded—or redeveloped, depending on the way one looks at it—version of several of its earlier versions. Our own little 3-tonne liquid engine was a 'grown-up' version of our 600 kg engine.

Now SEP conceptualized an upgraded version of M-40, with a 50-tonne thrust, but they did not call it M-50. Instead, they named it Viking-1. This, still, wasn't good enough. Then came Viking-2 as a 55-tonne engine, and Viking-3, a 60 tonner. It was a cluster of four such engines which went on to power the first stage of the first Ariane flight in 1977. Ariane used a Viking-4 engine, a slightly modified version of Viking-3 with an extended nozzle that would yield 73-tonne thrust in upper atmosphere, as the second stage, and the original HM-7 cryogenic engine as the upper stage.

Jointly developed by SEP and ISRO, Viking-3—Vikas for Indian rockets—continues to be the second stage of PSLV, India's most consistently successful launch vehicle. But how the joint venture came about is another tale of intense lobbying by friends like Grover, some smart deal-making by me and a crafty contract writing by T N Seshan, who was the administrative director of ISRO.